

Daniel Scalena

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Education

Double Doctorate (Ph.D) in Computer Science - NLP

Nov 2023 – exp. Q4 2026

University of Milano - Bicocca | University of Groningen (CLCG)

Researching interpretability to improve the safety, reliability, and real-world applicability of generative models.

Mediterranean Machine Learning (M2L) Summer School

Sep 2024

Milano, Italy

1-week program led by Google DeepMind scientists; presented my [paper](#) at BlackBoxNLP.

M.Sc in Computer Science, ML track

Oct 2021 – Oct 2023

University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 29.7/30)
- Thesis: On the explainability of Large Language Models detoxification.

B.Sc in Computer Science

Oct 2018 – Jul 2021

University of Milano - Bicocca

- Final Grade: 110/110 with honors, (GPA: 28.3/30)
- Thesis: Hate speech detection on social networks using state-of-the-art NLP techniques.

Experience

Cohere

June 2026 – Sep 2026

Intern – Member of Technical Staff

- Conducting research on reward design for code generation with the Code Agents team;
- Investigating RL and inference-time scaling approaches to improve agent reliability.

University of Groningen, Computational Linguistic Group

Apr 2023 – Jul 2023

Research Intern

- Aligned LMs to counter hate speech narratives using RLAIIF, reaching up to 50% in safety improvements;
- Analyzed and interpreted model behavior shifts after applying post-training algorithms.

C.I.N.I. & University of Milano - Bicocca

Jun 2022 – Apr 2023

Assistant Researcher

- Devised Open Relation Extraction for Italian legal texts, improving previous accuracy by 30%;
- Project funded by the *Italian Ministry of Justice* and *Inter-university Consortium for Computer Science*.

TESTUDO s.r.l.

Jun 2022 – Aug 2022

ML Engineer (contractor)

- Designed and trained deep learning models to automate the company's recognition system.

Main publications

🔗 EAGer: Entropy-Aware GENEration for Adaptive Inference-Time Scaling

Daniel Scalena, Leonidas Zotos, Elisabetta Fersini, Malvina Nissim, Ahmet Üstün

Proceedings of the 43rd International Conference on Machine Learning, (ICML 2026)

🔗 Steering Large Language Models for Machine Translation Personalization

Daniel Scalena*, Gabriele Sarti*, Arianna Bisazza, Elisabetta Fersini, Malvina Nissim

Proceedings of the 19th Conference of the European Chapter of the Association for CL, (EACL 2026)

[🔗 Multi-property steering of large language models with dynamic activation composition](#)

Daniel Scalena, Gabriele Sarti, Malvina Nissim

Proceedings at the 7th BlackBoxNLP Workshop, (at EMNLP 2024)

[🔗 A gentle push funziona benissimo: making instructed models in Italian via contrastive activation steering](#)

Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini

Proceedings at the 10th Italian Conference on Computational Linguistics, (CLIC-it 2024)

[🔗 Let the Models Respond: Interpreting Language Model Detoxification Through the Lens of Prompt Dependence](#)

Daniel Scalena, Gabriele Sarti, Malvina Nissim, Elisabetta Fersini

Extended abstract at the 6th BlackboxNLP Workshop, (at EMNLP 2023)

[Full list of publications](#) [🔗](#)

Scholarships and Awards

Google and Apple - BlackBoxNLP (EMNLP 2023) travel award	2023
Full-Fee M.Sc. Scholarship due to merit and financial scores	2021 - 2023
Full-Fee B.Sc. Scholarship due to merit and financial scores	2018 - 2021

Projects

Reward LM

[DanielSc4/RewardLM](#) [🔗](#)

- Python toolkit library for Fine-Tuning and Reinforcement Learning from AI Feedback of Generative LMs;
- Developed tools to interpret and compare toxicity differences between original and fine-tuned models.

Activities

Co-organiser	Challenge the Abilities of LAngeuage Models in ITAliaN (CALAMITA) 2024
Teaching	NLP course labs, MSc Data Science @ UniMiB (2024 - 2026)
Reviewer	NeurIPS 2026, ICML 2026, EMNLP 2025 - 2026, BlackBox-NLP 2025, ACL 2025, LREC-Coling 2024, NAACL 2024*, Information Processing and Management 2024 - 2025, CliC-IT 2023 - 2026, ICML Mechanistic Interpretability Workshop 2024* (*secondary reviewer)
Volunteer	CliC-IT 2021, EMNLP 2023
Misc.	European Researchers Night GroNLP collaborator 2023

Languages

Italian	Native Language
English	C1 - B-between Foreign Languages 🔗
French	B1